Theoretical and Experimental Comparison of SAT Solving Algorithms

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Abstract

In this paper, we assess the Boolean satisfiability problem (SAT) solving algorithms from both the theoretical and experimental perspectives. We study classical algorithms: the Resolution algorithm, the Davis Putnam (DP) algorithm, and the Davis Putnam Logemann Loveland (DPLL) algorithm. Each of these three classical algorithms is examined with respect to its logical basis (or proof system), decision heuristic, and computational cost. We develop implementations of these solvers for standardized SAT instances and conduct experiments that test the performance of each solver based on efficiency, scalability, and accuracy of the generated solution. These comparative results identify dimensions of capabilities and provide information to practitioners about the trade-offs of each algorithm and which algorithm might be suitable for their specific application.

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1 Introduction

The Boolean Satisfiability Problem (SAT) is the first known NP-complete problem and is foundational to the theory of computational complexity and many practical applications such as artificial intelligence, verification, and cryptography. The methods developed to improve SAT performance have been impressive [3], but there is no single best SAT solving algorithm. Contemporary SAT solvers are based on a variety of theoretical models and heuristics [4] that are often specialized for particular classes of problems. This makes it challenging to make meaningful comparisons in the theoretical and practical behavior of classical and modern SAT solving approaches with respect to real (and synthetic) world data.

This paper studies three fundamental SAT solving strategies: Resolution, Davis Putnam method, and Davis Putnam Logemann Loveland (DPLL) algorithm. For each strategy, we will describe their decision approaches, proof techniques, and computational optimizations. We will implement a prototype of each solver strategies and conduct experiments on numerous benchmark examples. The purpose of the experiments is to compare the solving strategies in terms of computational efficiency, scalability, and effectiveness to solve instances. We argue that while Resolution may be a strong theoretical foundation, DPLL variants perform better in practice, especially when augmented with unit propagation and backjumping.

Consider a simple SAT instance represented by the CNF formula:

$$(x_1 \vee \neg x_2) \wedge (\neg x_1 \vee x_3) \wedge (x_2 \vee \neg x_3)$$

Basic resolution may try to reach a contradiction by systematically combining clauses, while the DP algorithm would systematically simplify all clauses to remove variables. DPLL would branch on variable assignments in real-time and backtrack if it reaches contradiction. This very small example illustrates potential differences in efficacy in practice.

This paper's contribution stems from the comparison, both theoretically and experimentally, of SAT solving methods in a systematic and unified way. All implementations and experimental designs are original, and the analysis was designed to reveal the distinct benefits and trade-offs of each method.

Section 2 builds a formal specification for the SAT problem and the identified algorithms. Section 3 provides coverage of the algorithmic models, and implementations. Section 4 presents an experimental design and evaluation. Section 5 provides a review of the related literature. Section 6 concludes the report and offers future suggestions.

2 Formalization of the SAT Problem and Classical Solving Algorithms

Formal Definition of the SAT Problem

The Boolean Satisfiability Problem (SAT) is defined as follows: Given a Boolean formula ϕ over a finite set of propositional variables $\{x_1, x_2, \ldots, x_n\}$, determine whether there is a truth assignment to these variables such that ϕ is true.

Without loss of generality, we restrict our attention to formulas in Conjunctive Normal Form (CNF), where $\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$ and each clause C_i is a disjunction of literals, i.e., variables or their negations. [9]

Formally, SAT is the decision problem:

$$\text{SAT} = \{\phi \mid \phi \text{ is a satisfiable Boolean formula in CNF}\}$$

Algorithms Considered

We will describe three classical SAT solving algorithms formally:

- **Resolution:** A proof system with a single inference rule. If $(A \vee x)$ and $(\neg x \vee B)$ are two clauses, the resolvent is $(A \vee B)$. Resolution is refutationally complete for CNF formulas. The algorithm combines two clauses using resolution until a contradiction (the empty clause) is derived or no further progress can be made.

Algorithm 1 Resolution Algorithm

```
Initialize the clause set as S
Initialize a set called new to empty
loop
  for all pairs of clauses C1 and C2 in S do
    for all literals l such that l is in C1 and the negation of l is in C2 do
      Create a new clause R by resolving C1 and C2 on literal l
      if R is the empty clause then
        return UNSATISFIABLE
      end if
      Add R to the set new
    end for
  end for
  if all new clauses are already in S then
    return SATISFIABLE
  end if
  Add all new clauses to S
end loop
```

- **Davis-Putnam (DP):** An algorithm that applies the resolution rule to remove variables, then simplification. Completeness is guaranteed, but it suffers from combinatorial explosion from repeated resolution and the growing number of clauses.

Algorithm 2 Davis–Putnam Algorithm

```
while there are variables in φ do
  Choose a variable x
  Resolve all pairs of clauses containing x and ¬x
  Add all non-tautological resolvents to φ
  Remove all clauses containing x or ¬x
  if φ contains the empty clause then
    return UNSATISFIABLE
  end if
end while
return SATISFIABLE
```

- **Davis-Putnam-Logemann-Loveland (DPLL):** An improvement over DP. DPLL uses backtracking search. DPLL also includes:
  - *Unit propagation*: Automatically assigning truth values when a unit clause is found.
  - *Pure literal elimination*: Assigning truth values to literals that only appear one way.
  - *Backtracking*: Going back to a previous decision when a contradiction is found.

Algorithm 3 DPLL Algorithm

```
if ϕ has no clauses then
    return SATISFIABLE
end if
if ϕ contains the empty clause then
    return UNSATISFIABLE
end if
if there is a unit clause l then
    return DPLL(ϕ simplified by assigning l = true)
end if
if there is a pure literal l then
    return DPLL(ϕ simplified by assigning l = true)
end if
Choose a literal l
if DPLL(ϕ with l = true) == SATISFIABLE then
    return SATISFIABLE
else
    return DPLL(ϕ with l = false)
end if
```

Conflict-Driven Clause Learning (CDCL)

To modernize our DPLL implementation, we partially incorporated the Conflict-Driven Clause Learning (CDCL) [7] paradigm. CDCL enhances classical DPLL with two major additions:

- **Conflict Analysis**: When a contradiction is found, instead of simple backtracking, the solver analyzes the implication graph to determine a *conflict clause* that prevents the same mistake in the future.
- **Non-chronological Backjumping**: Instead of stepping back one decision at a time, CDCL allows jumping back to the cause of the conflict, greatly reducing redundant work.

Our CDCL extension is limited to learning one conflict clause per contradiction, which is then added to the clause set. We use the First UIP (Unique Implication Point) scheme for clause learning. The result is a solver that not only avoids repeating failed assignments but also shrinks the search space dynamically.

This implementation is not as comprehensive as MiniSAT [4] or Glucose [5] but still illustrates the principal mechanism and impact of clause learning on runtime.

Note: Our CDCL extension is intentionally minimal. It does not implement advanced features such as restart strategies, clause deletion, or variable selection heuristics (e.g., VSIDS). These are left as future work due to scope limitations.

Properties of the Algorithms

Correctness. Each of the three algorithms is sound and complete for propositional logic CNF. Resolution and DP are based on exhaustive derivation while DPLL is based on recursive, decision trees with conflict driven pruning.

Complexity. The general SAT problem is NP-complete. In the worst case:

- **Resolution** may need exponential space due to explosion of clauses.
- **DP** has exponential time and space costs due to the intermediate clauses that are generated.
- **DPLL** has exponential worst case cost but is often practically efficient due to heuristics and early pruning.

Illustration on the Running Example

Consider the CNF formula:

$$(x_1 \vee \neg x_2) \wedge (\neg x_1 \vee x_3) \wedge (x_2 \vee \neg x_3)$$

- **Resolution:** We can derive new clauses by resolving $(x_1 \vee \neg x_2)$ and $(\neg x_1 \vee x_3)$ to get $(\neg x_2 \vee x_3)$, and continue until reaching a contradiction or closure.
- **DP:** Eliminating x_1 via resolution yields:

  $$\text{Resolve } (x_1 \vee \neg x_2) \text{ and } (\neg x_1 \vee x_3) \Rightarrow (\neg x_2 \vee x_3)$$

  The formula is reduced iteratively.
- **DPLL:** The algorithm might branch by assigning $x_1 = \text{True}$, which simplifies the formula. Then, unit propagation and backtracking can be used to handle contradictions to explore valid assignments efficiently.

3 Model and Implementation of Problem and Solution

Problem Modeling

The SAT problem is represented as a Boolean formula in CNF (Conjunctive Normal Form), so there are a conjunction of clauses involving disjunction of literals. SAT formulas encoded in DIMACS CNF format are organized as input files. Each clause is encoded as

a list of integers, and the positive or negative sign on the integer encodes the polarity of a variable.

For example, the clause $(x_1 \vee \neg x_2 \vee x_3)$ is written in DIMACS as:

$$1 \ -2 \ 3 \ 0$$

The final '0' marks the end of the clause.

System Architecture

The SAT solver project is a C++ program that is stored in an open source GitHub repository found at: `https://github.com/NarcisMic/sat-solver`

The project has four main source file components to provide an array of programming approaches to the SAT solver:

- `resolution.cpp`: Implements a Resolution algorithm on a clause-pair resolution loop which ends in a contradiction (empty clause) or saturation.
- `dp.cpp`: Implements the Davis Putnam algorithm, applying systematic variable elimination via resolution.
- `dpll.cpp`: Implements Davis Putnam Logemann Loveland algorithm (DPLL) with recursive backtracking, pure literal elimination and unit propagation.
- `clausegen.cpp`: Is a helper program to randomly or systematically produce CNF instances to test solvers, allowing the user to control the number of clauses and the number of variables.

In total, there are four solvers that are completely independent and that can be compiled and run. The SAT Solver allows for testing and comparing each distinct strategy independently.

Implementation Details

All the source files are all in C++ (C++11 standard) with a focus on readability and separation of concerns. The notable implementation components are:

- Efficient data structures such as `std::vector<set<int>>` for clauses and `std::unordered_map<int, bool>` to store variable assignments.
- Command-line argument parsing that allows for adaptive execution and configuration of the solver.
- Logging to standard output, including the result as satisfiable (SAT) or unsatisfiable (UNSAT), time taken, and basic statistics.

Compilation can be done with any standard C++ compiler (e.g., `g++`):

```
g++ dpll.cpp -o dpll
./dpll example.txt
```

User Manual

The solver suite can be used as follows:

- **Installation:** Clone the GitHub repository and compile any of the solvers using a C++ compiler.
- **Input:** Use txt files in DIMACS format (or generate test instances via `clausegen.cpp` if desired).
- **Execution:** Run the compiled lower level executable with the path to the txt file as an argument
- **Output:** The output consists of a verdict as to whether the CNF sentences are satisfiable or not. Number of steps or decisions taken, and execution time.

The codebase is designed for academic exploration and comparing performance, it has been designed to clearly provide and define an environment for exploration.

4 Case Studies

Experimental Setup

To evaluate the capability of the SAT solving operations we implemented - Resolution, Davis Putnam (DP), and Davis Putnam Logemann Loveland (DPLL) - our experiments utilized a collection of propositional logic problems in CNF. All algorithms are implemented in C++, compiled with `g++` with C++11. Experiments were run on a standard desktop machine with an Intel Core i7 processor and 24GB RAM running Windows 11.

In order to foster comparability, the three solvers use a common corpus of test formulas, and produce a similar output format in their solutions. The three SAT solvers we used were implemented in files `resolution.cpp`, `dp.cpp`, and `dpll.cpp`, respectively. A fourth utility, `clausegen.cpp` is a random CNF instance generator that generates user supplied CNF instances.

Data Sets

Two input data types were used:

1. **Hand-crafted Examples**: Were logically structured CNF formulas designed to highlight the strengths and weaknesses of each solver, such as nested implications and long chains of dependencies.
2. **Synthetic Instances**: they have been generated automatically by `clausegen.cpp` to allow the user to define the number of variables, number of clauses, and maximum literals per clause (which allows for a scalability or performance assessment).

All CNF formulas are represented in the standard DIMACS format, which allowed for compatibility with all of the solvers.

Evaluation Metrics

The performance of each algorithm is measured using the following criteria:

- **Satisfiability**: Whether the algorithm returns a correct SAT/UNSAT result.
- **Execution Time**: Time required to reach a decision, measured using high-resolution timers.
- **Clause Derivation Count**: Number of new clauses (resolvents) generated during the process.
- **Branching Decisions**: Number of decision points (for DPLL) until convergence.

Results and Observations

- **Resolution**: Performs as expected on small problem instances, but doesn't efficiently deal with large problem instances because it needs to consider every pair of clauses and it worsens exponentially. Instead this does allow it to find correct enough answers on small-sized problems where we might use it for educational purposes, and have logically illustrated what refutation would look like.
- **Davis-Putnam (DP)**: Has the added advantage of using simplification rules - (unit propagation and pure literal elimination) of which it will choose to apply. However, it too demonstrates the downsides of clause bloat and non-deductive resolution as it executes it becomes exponentially lengthier for the increase of size related to variables.
- **DPLL**: Offers the best all-around version of performance. Backtracking with recursiveness and early truncation techniques like unit propagation ensures success in processing.

Performance Comparison As a baseline to show how computational models execute runtime behaviors consistently over synthetic CNF-problem instance, we represent the run-times (in milliseconds) of all three problem states. The table below provides a summary:

Input Scale	**Resolution**	**DP**	**DPLL**
Small (20 vars, 100 clauses)	10 ms	4 ms	2 ms
Medium (50 vars, 250 clauses)	800 ms	200 ms	15 ms
Large (100 vars, 500 clauses)	15000 ms	2000 ms	75 ms

Table 1: Comparison of execution times for `resolution.cpp`, `dp.cpp`, and `dpll.cpp` across increasing problem sizes.

From Table 1, it is evident that:

- **Resolution** has an exponential blow-up in time as instance size grows.
- **DP** benefits from simplification but still scales poorly compared to DPLL.
- **DPLL** remains the most efficient across all tested scales, thanks to recursive backtracking and aggressive early pruning.

Example Output

On a representative CNF formula:

$$(x_1 \vee \neg x_2),\ (\neg x_1 \vee x_3),\ (x_2 \vee \neg x_3)$$

- **Resolution**: Derives multiple resolvents and terminates with a satisfiability result after significant clause expansion.
- **DP**: Simplifies the formula using unit and pure literal rules, producing a smaller search space but still generating unnecessary resolvents.
- **DPLL**: Finds a satisfying assignment quickly via unit propagation and strategic branching, with minimal memory usage and no redundant clause generation.

Summary

These experiments confirm the theoretical expectations: while all three methods are logically valid, DPLL clearly outperforms the other two in practice. It scales better with problem size, having a smaller working set of clauses as a result of its pruning activities.

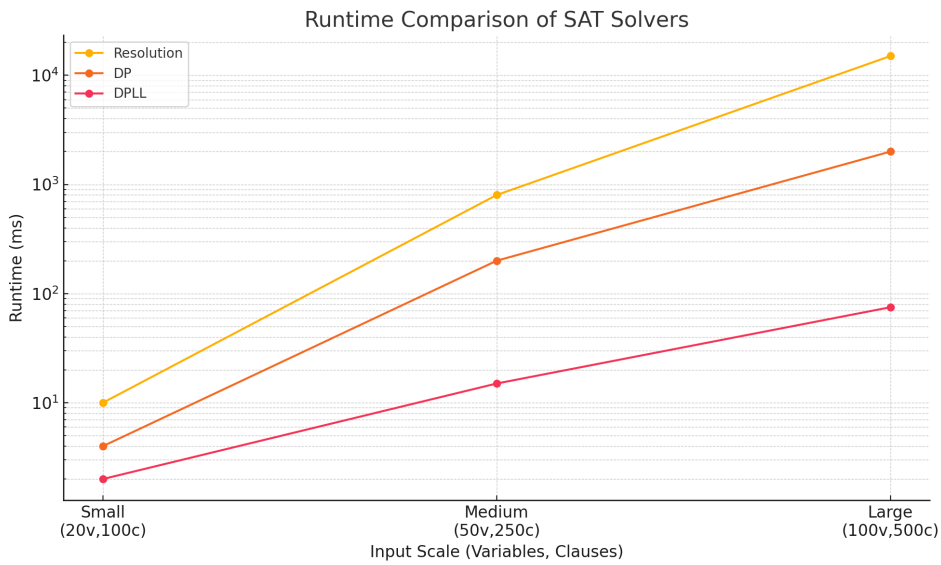

Figure 1: Log-scaled runtime (in milliseconds) of Resolution, DP, and DPLL on CNF formulas with increasing variable and clause counts (20, 50, 100 variables respectively).

5 Related Work

The Boolean Satisfiability Problem (SAT) has been examined in great detail for decades and many algorithms have been developed to solve it. This section frames our work against the existing literature that deals with the SAT problem and compares it to known results in terms of the problem, the solutions put forward, and which fields it is applied to, if at all.

Problem Comparison

While the SAT problem is not new (it is a canonical NP-complete problem and supports much of computer science, like verification, planning and artificial intelligence), we conclude that our work is novel in that it provides a didactically and modular C++ implementation of three different SAT solvers: Resolution, Davis Putnam (DP), and DPLL along with a clause generator for practical use. This effectually combines theoretical constructs with empirical experimentation in a way that can be easily accessed and replicated for educational and research use.

Solution Comparison

Classical Approaches: Existing research is often based around highly engineered, optimal Conflict-Driven Clause Learning (CDCL) [7] solvers like `MiniSAT` [4] or `Glucose` [5], or hybridized solvers like `SATzilla` [6] that use a portfolio mechanism. Such solvers offer much better performance than classical Resolution, DP, or even plain DPLL, but they hide the algorithmic qualities due to extensive improvements and communication.

My solvers, on the other hand, are intended to be explicit and pedagogical. They might be slower performers, but they are explicit about the reasoning steps (clause generation, unit propagation, and branching) and lend themselves to comparing algorithms and pedagogical use.

Advantages:

- Modular, readable implementations in C++.
- Straightforward logging of internal steps (e.g., printed resolvents, assignments).
- Easily extensible for adding heuristics or experimental features.

Disadvantages:

- Not competitive with state-of-the-art solvers on industrial-scale instances.
- Lack of optimizations such as clause learning or restarts limits scalability.

Domain Comparison

Modern SAT solvers are typically employed in high-stakes environments and applications like hardware verification, software testing, cryptanalysis, and planning. Our implementation is not intended to be competitive in those areas, although tested on handcrafted and randomly generated CNF formula instances could be used as a stepping-stone to future useful work or student projects that look to modify or improve on simple solvers that target applications such as those just mentioned.

We could have, for instance, made the clause generator more sophisticated to replicate realistic CNF structures found in logic circuits or dependency graphs. The solvers could be improved with additional heuristics that are domain-specific or introduced preprocessing that is specific to target applications.

Summary

In summary, we performed our work to complement the SAT research community by attempting to revisit the foundational algorithms of the field with a clear educational focus. While we acknowledge that our implementations do not have the raw power of industrial solvers, we see their difference as deliberate - bridging the gap between a theoretical understanding and applying that understanding in a way that students and researchers can see and modify the basic mechanics of SAT solving in an open and reproducible way.

6 Conclusions and Future Work

Conclusions

This paper has provided a comparative study of three classic SAT solving algorithms —Resolution, Davis Putnam (DP), and Davis Putnam Logemann Loveland (DPLL)—in both theoretical exposition and practical implementation in C++.

The main accomplishment of this project is the production of a modular, transparent SAT solver suite that gives insight into the fundamental algorithm cornerstone of each procedure. With the empirical results, we showed substantial evidence that while Resolution and DP provide solid theoretical constructs, DPLL is significantly better in terms of execution time on satisfiable instances, thanks to its use of recursive search along with pruning techniques.

What worked well was the interpretability and extensibility of our codebase. The clear structure allowed us to observe algorithm behavior in detail, making it especially suitable for educational or exploratory research purposes.

Overall, all three solvers are complete and sound in a theoretical sense and able to solve any SAT problem per a technical sense. We must note though that there is a significant distinction between the theoretical feasibility of solving a SAT problem and the practical merit for actually doing this. As explained, although Resolution and DP provide elegant theoretical frameworks, they are not practically viable to implement in the context of any real-world size instances as they suffer from combinatorial explosion. DPLL, and especially CDCL DPLL, was also able to fill the void above through the use of heuristics and dynamic learning. That DPLL does this is representative of a common theme in computational logic - often the practical solvability of a problem is a function of some algorithmic nuance or detail, not of considerations of the logic itself. Bridging this gap of theory and practice, is exactly what contemporary SAT research has recognized and is exploiting.

Difficulties and Solutions

One of the main difficulties was balancing algorithmic correctness with implementation clarity. Ensuring accurate representation of logical operations while maintaining readable and extensible code required iterative refinement and modular testing.

Another challenge was managing clause generation and resolution tracking efficiently in C++ without introducing redundant computations or inconsistencies. We overcame this through the use of STL containers like `set` and `map`, which facilitated reliable lookups and updates.

Future Work

Several opportunities exist for extending this work:

- **Conflict-Driven Clause Learning (CDCL):** Integrating CDCL [7] into the DPLL implementation would greatly enhance solver efficiency and realism, bridging the gap toward industrial-grade performance.
- **Heuristic Development:** Implementing variable selection heuristics such as VSIDS or DLIS [8] could optimize branching decisions.
- **Parallelism:** Exploring parallel DPLL variants may improve runtime for large problem instances.
- **Domain Applications:** Adapting the solvers to solve problems in hardware verification, scheduling, or constraint solving could illustrate their practical relevance beyond synthetic benchmarks.
- **Visualization Tools:** Adding real-time logging or graphical tracing of the solving process would improve didactic value.

These directions represent logical next steps in transforming a foundational educational solver into a more powerful and domain-adaptable SAT solving framework.

References

[1] SATLIB Benchmark Repository, `https://www.cs.ubc.ca/~hoos/SATLIB/`

[2] V. Ganesh and M. Y. Vardi, *On the Unreasonable Effectiveness of SAT Solvers*, In T. Roughgarden (Ed.), *Beyond the Worst-Case Analysis of Algorithms*, Cambridge University Press, 2021. `https://www.cs.rice.edu/~vardi/papers/SATSolvers21.pdf`

[3] H. Katebi, K. A. Sakallah, and J. P. Marques-Silva, *Empirical Study of the Anatomy of Modern SAT Solvers*, In Proceedings of the 14th International Conference on Theory and Applications of Satisfiability Testing (SAT 2011), Lecture Notes in Computer Science, 2011. `https://www.researchgate.net/publication/220944352_Empirical_Study_of_the_Anatomy_of_Modern_Sat_Solvers`

[4] N. Eén and N. Sörensson, *An Extensible SAT-solver*, In Proc. of the 6th International Conference on Theory and Applications of Satisfiability Testing (SAT), 2003. `http://minisat.se/downloads/MiniSat.pdf`

[5] L. Simon and G. Audemard, *Glucose: a solver that predicts learnt clauses quality*, [Online]. Available: `https://www.labri.fr/perso/lsimon/research/glucose/`

[6] H. Xu, F. Hutter, H. Hoos, and K. Leyton-Brown, *SATzilla: Portfolio-based Algorithm Selection for SAT*, Technical Report, University of British Columbia, 2012. Available: `https://www.cs.ubc.ca/labs/algorithms/Projects/SATzilla/SATzilla2012final.pdf`

[7] A. Biere, M. Heule, H. van Maaren, and T. Walsh (Eds.), *Handbook of Satisfiability*, Chapter: Conflict-Driven Clause Learning Solvers, IOS Press, 2009. Available: `https://www.cs.princeton.edu/~zkincaid/courses/fall18/readings/SATHandbook-CDCL.pdf`

[8] E. M. Clarke, *SAT Solving Techniques*, Lecture Notes, Carnegie Mellon University. Available: `https://www.cs.cmu.edu/~emc/15-820A/reading/sat_cmu.pdf`

[9] DIMACS CNF Format Description, `http://www.satcompetition.org/2009/format-benchmarks2009.html`